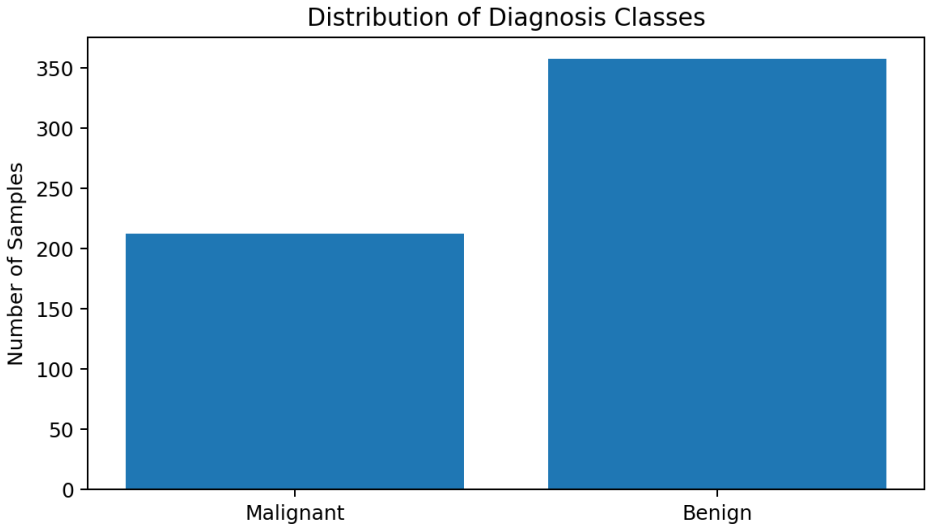
BREAST CANCER CLASSIFICATION USING MACHINE LEARNING

An ML case study using the Wisconsin Diagnostic dataset



Models evaluated

Logistic Regression • Support Vector Machine (SVM) • Decision Tree • Random Forest

Prepared from the Google Colab implementation and its recorded outputs

Abstract

This case study evaluates four supervised machine-learning classifiers for distinguishing benign from malignant breast-mass samples using numerical measurements computed from digitized fine-needle aspirate images. The workflow includes exploratory data analysis, an 80:20 stratified train-test split, standardization, principal component analysis (PCA), model comparison, confusion matrices, ROC analysis and Random Forest feature importance. On the held-out test set of 114 samples, Logistic Regression and SVM achieved 98.25% accuracy using all 30 standardized features. Logistic Regression had the highest reported ROC-AUC (0.9954). PCA reduced the feature representation to 10 components while retaining 95.27% of training-set variance; performance changed slightly depending on the model. These are results from one fixed split and should be interpreted as an educational experiment, not as evidence of clinical readiness.

Keywords: breast cancer classification, machine learning, Logistic Regression, SVM, Random Forest, PCA, ROC-AUC

Report Structure

1. Introduction
2. Objectives
3. Dataset Description
4. Methodology
5. Exploratory Data Analysis
6. Preprocessing and PCA
7. Models and Evaluation Metrics
8. Results and Discussion
9. Limitations and Future Scope
10. Conclusion

1. Introduction

Breast-mass classification is a binary classification problem in which a sample is labelled benign or malignant. Machine-learning methods can learn patterns in numerical measurements and produce predictions for evaluation. In this case study, the supplied dataset is used to compare several standard classifiers under the same train-test split. The aim is to study model behaviour and evaluation methods, rather than to create a diagnostic tool for clinical use.

2. Objectives

• Explore the dataset and understand the distribution of the two diagnosis classes.
• Prepare the numerical features for model training using a stratified split and standardization.
• Compare Logistic Regression, SVM, Decision Tree and Random Forest.
• Study dimensionality reduction using PCA while retaining at least 95% of training variance.
• Evaluate models using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.
• Inspect Random Forest feature importance and summarize the limitations of the experiment.

3. Dataset Description

The notebook uses Scikit-learn's built-in Breast Cancer Wisconsin (Diagnostic) dataset through `load_breast_cancer(as_frame=True)`. The data is bundled with Scikit-learn, so a separate file upload is not required for this implementation.

Property	Description
Number of samples	569
Input features	30 numerical features
Target classes	Malignant and benign
Benign samples	357
Malignant samples	212
Missing values	0
Feature data type	Floating-point numerical values

The features describe cell-nucleus measurements such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension. For many properties, the dataset contains mean, error and worst-value measurements.

Target encoding used in the notebook

Scikit-learn encodes target 0 as malignant and target 1 as benign. The report keeps this mapping consistent. Some evaluation tables generated by the notebook use the default positive class (label 1, benign); therefore, malignant-specific precision and recall are also reported separately.

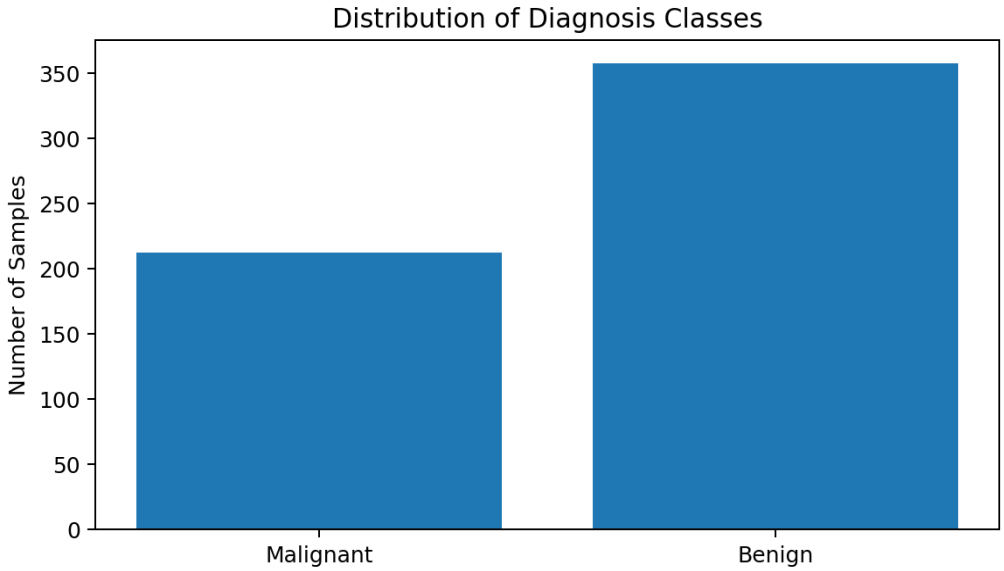


Figure 1. Class distribution: 357 benign and 212 malignant samples.

4. Methodology

The experiment follows this sequence: load data → explore data → split into training and testing sets → standardize features → fit PCA on training data → train four classifiers on original features → evaluate predictions → train the same classifiers on PCA features → compare results → examine Random Forest feature importance.

The dataset was divided using an 80:20 stratified split with random_state = 42. This produced 455 training samples and 114 test samples. Stratification preserves approximately the same class proportions in both sets.

5. Exploratory Data Analysis (EDA)

5.1 Dataset quality and summary

The notebook reports 569 rows and 30 input columns, all numerical. No missing values were detected. Therefore, missing-value imputation was not required for this dataset.

5.2 Feature distributions

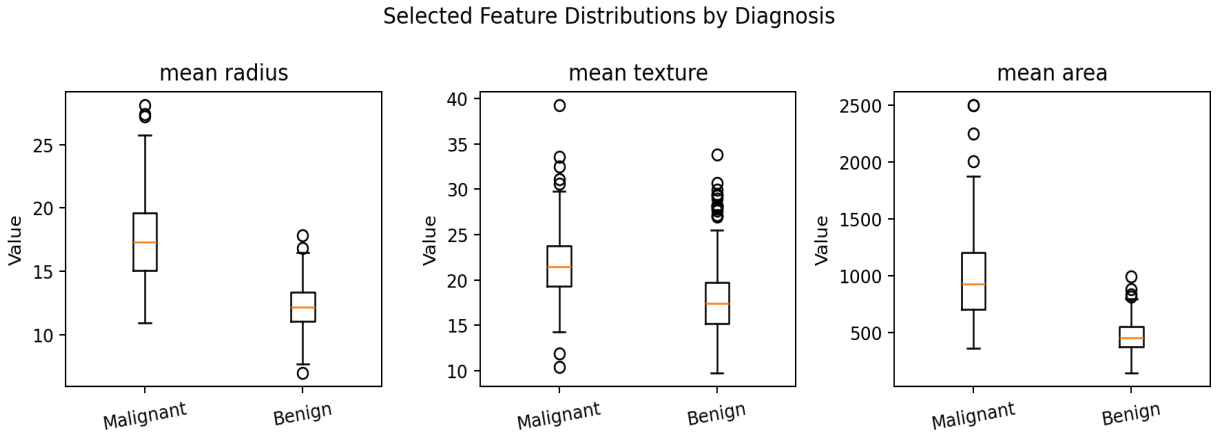


Figure 2. Boxplots of mean radius, mean texture and mean area by diagnosis.

The boxplots allow visual comparison of the median, spread and overlap for each selected feature. In the observed plots, mean radius and mean area tend to be higher for malignant samples, while overlap remains between classes. These visual patterns suggest potential predictive value but do not alone establish classification performance.

5.3 Correlation analysis

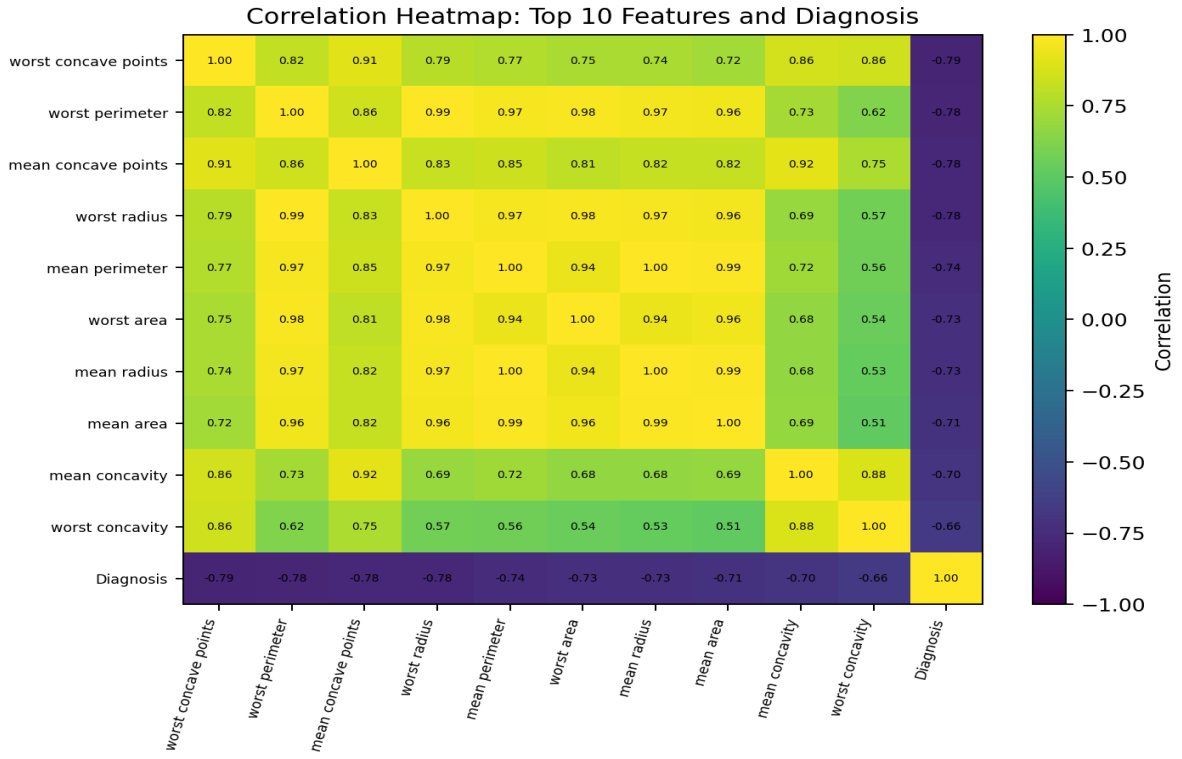


Figure 3. Correlation heatmap for the ten features with the largest absolute correlation with the encoded diagnosis target.

The strongest absolute target correlation reported by the notebook was for worst concave points (0.794), followed by worst perimeter (0.783), mean concave points (0.777) and worst radius (0.776). Several size-related features are also strongly correlated with one another. Because the target encoding is 0 = malignant and 1 = benign, the sign of a correlation determines its direction; the absolute values used to select the top features do not show direction.

6. Preprocessing and Dimensionality Reduction

6.1 Train-test split and standardization

Item	Result
Training samples	455
Testing samples	114
Training feature matrix	455 × 30
Testing feature matrix	114 × 30
Training classes	285 benign; 170 malignant
Testing classes	72 benign; 42 malignant

StandardScaler was fitted only on the training data. The learned scaling parameters were then applied to the test data using transform(). This prevents test-set statistics from being used to fit the scaler.

6.2 Principal Component Analysis (PCA)

PCA was fitted on the standardized training features. The cumulative explained-variance curve indicated that 10 principal components were needed to retain at least 95% of the training-set variance. The reduced matrices had 10 columns instead of the original 30.

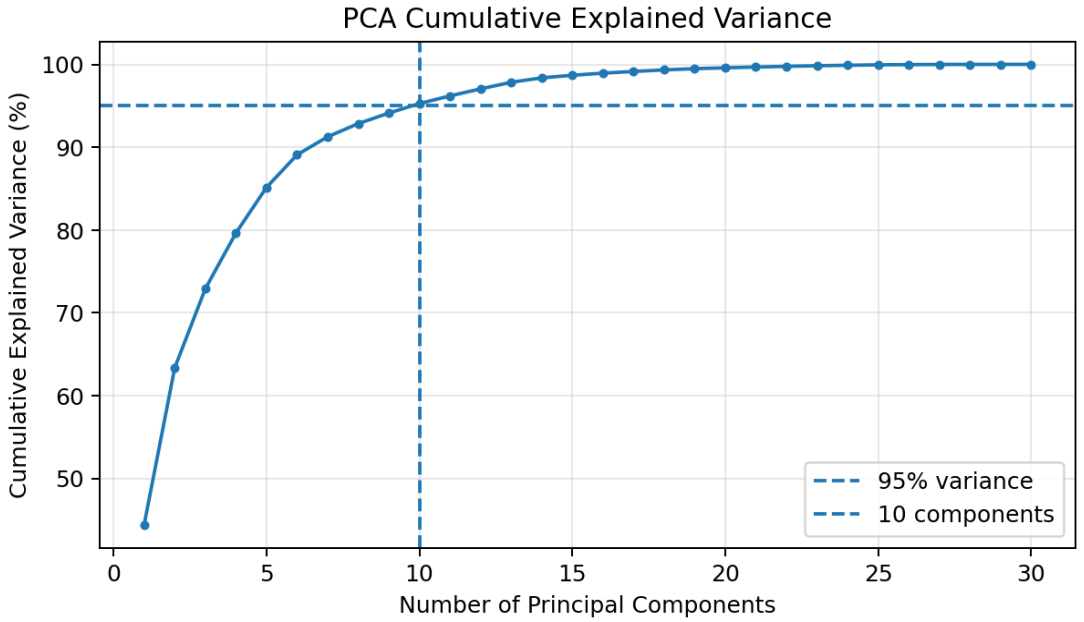


Figure 4. Cumulative explained variance; 10 components retain 95.27% of variance.

PCA property	Result
Original feature count	30
Components retained	10
Variance retained	95.27%
Dimensionality reduction	66.7% fewer input dimensions

PCA reduces dimensionality by combining correlated variables into components. Retaining 95.27% of total variance does not guarantee retaining 95.27% of class-discriminative information; classification performance must be measured separately.

7. Machine-Learning Models and Evaluation Metrics

Four classifiers were trained using the original 30 standardized features. The same four model types were also trained using the 10 PCA components.

7.1 Algorithms

Logistic Regression: A supervised classification algorithm that models class probability using a logistic function. The notebook used max_iter = 5000 and random_state = 42.

Support Vector Machine (SVM): A classifier that finds a separating boundary; the implementation used an RBF kernel and probability estimates.

Decision Tree: A tree-based classifier that learns feature-based decision rules. The random state was fixed at 42.

Random Forest: An ensemble of decision trees. The implementation used 100 trees and random_state = 42.

7.2 Evaluation metrics

Metric	Meaning
Accuracy	Proportion of all test samples classified correctly.
Precision	Among samples predicted as the positive class, the proportion that are truly positive.
Recall / Sensitivity	Among actual positive samples, the proportion correctly detected.
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Measures ranking/discrimination across classification thresholds.
Confusion matrix	Counts correct and incorrect predictions by actual and predicted class.

In the initial model-comparison tables, Scikit-learn's default positive label is 1 (benign). A separate evaluation treats malignant as the positive class so that malignant sensitivity is explicit.

8. Results and Discussion

8.1 Models using the original 30 features

Model	Accuracy	Precision*	Recall*	F1-score*	ROC-AUC*
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
SVM	0.9825	0.9861	0.9861	0.9861	0.9950
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9939
Decision Tree	0.9123	0.9559	0.9028	0.9286	0.9157

*Precision, recall, F1-score and ROC-AUC in this table follow the notebook's positive class label 1 = benign.

Logistic Regression and SVM achieved the highest accuracy (98.25%) and F1-score (0.9861). Logistic Regression had the highest reported ROC-AUC (0.9954), narrowly above SVM (0.9950). Random Forest achieved 95.61% accuracy, while Decision Tree achieved 91.23%.

8.2 Confusion matrices

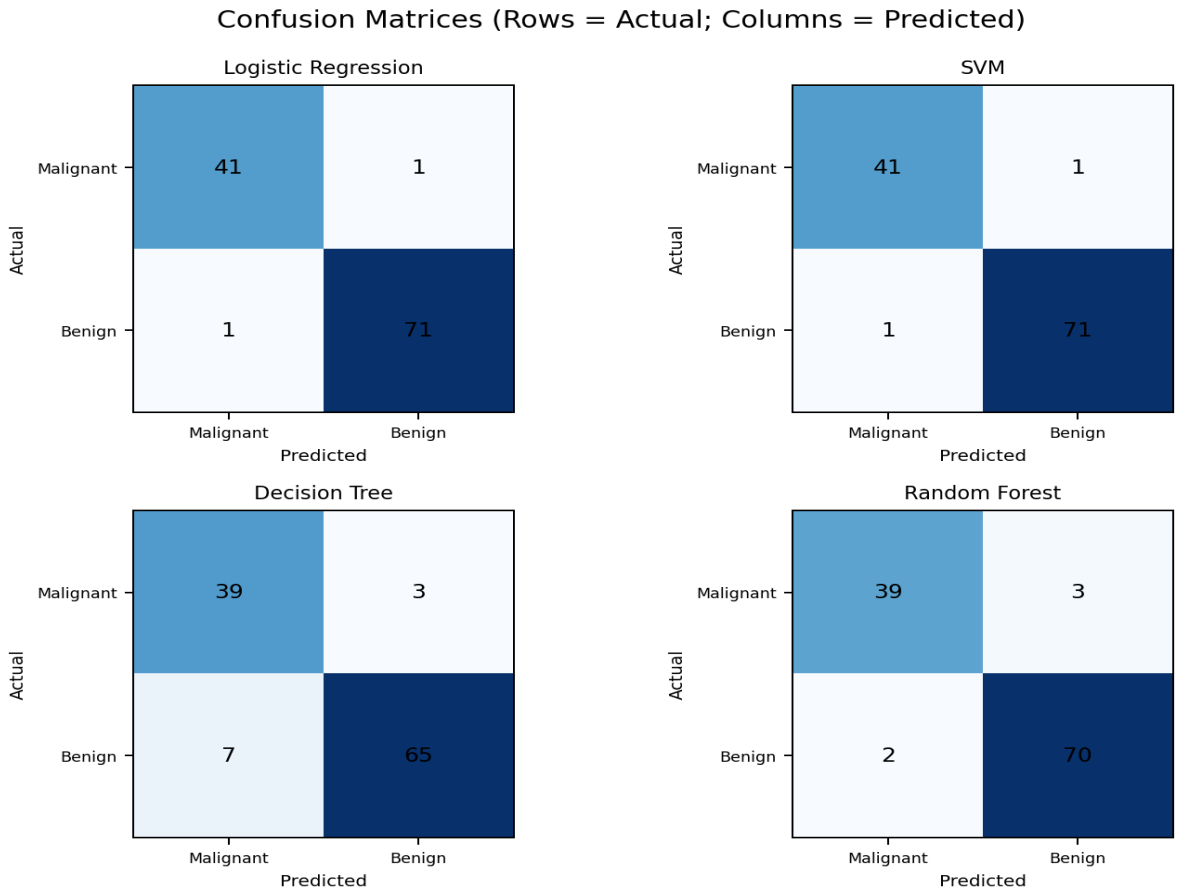


Figure 5. Confusion matrices. Rows show actual diagnosis; columns show predicted diagnosis. Class order is malignant, then benign.

For Logistic Regression and SVM, 41 of 42 malignant samples and 71 of 72 benign samples were classified correctly. Decision Tree correctly classified 39 malignant and 65 benign samples. Random Forest correctly classified 39 malignant and 70 benign samples.

8.3 ROC curve analysis

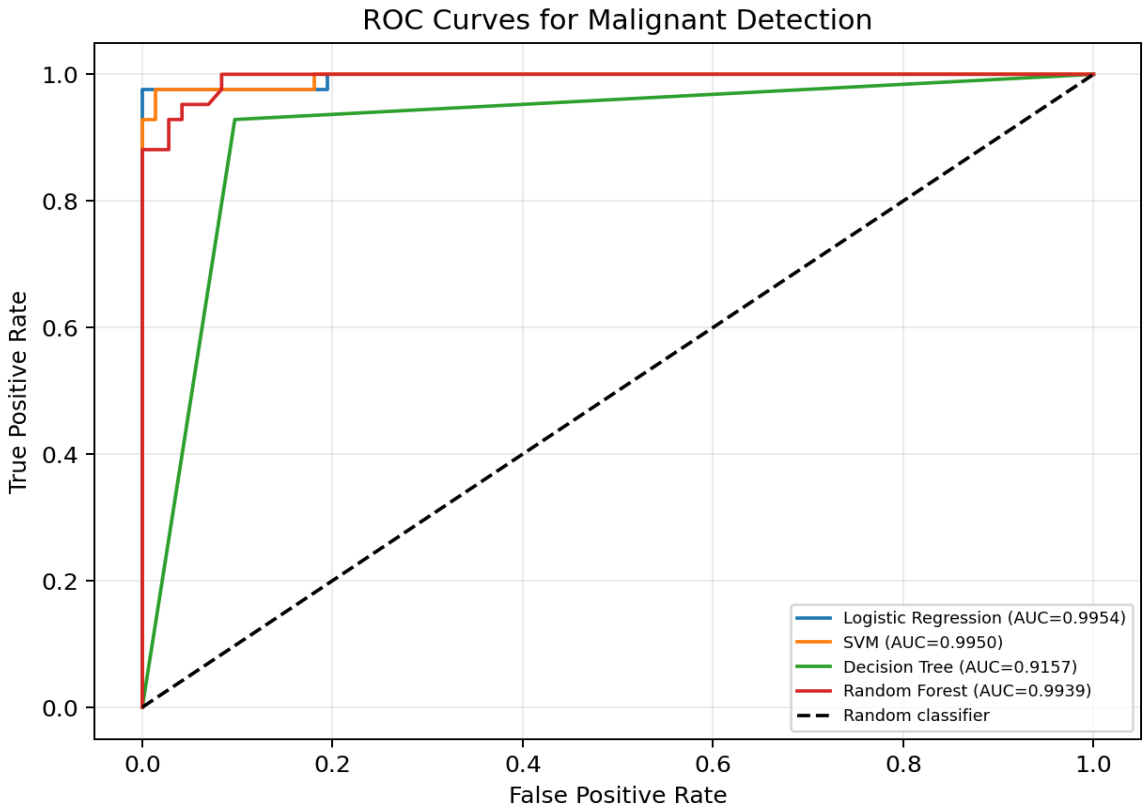


Figure 6. ROC curves computed using malignant-class probabilities as the positive class.

The ROC curves show strong ranking performance for Logistic Regression, SVM and Random Forest on this test split. The malignant-class AUC values are 0.9954 for Logistic Regression, 0.9950 for SVM, 0.9939 for Random Forest and 0.9157 for Decision Tree. AUC measures discrimination over thresholds; it does not by itself establish clinical usefulness.

8.4 Malignant-specific evaluation

Model	Accuracy	Malignant precision	Malignant recall	Malignant F1
Logistic Regression	0.9825	0.9762	0.9762	0.9762
SVM	0.9825	0.9762	0.9762	0.9762
Decision Tree	0.9123	0.8478	0.9286	0.8864
Random Forest	0.9561	0.9512	0.9286	0.9398

Logistic Regression and SVM achieved malignant recall of 0.9762 (41 of 42 malignant samples detected). Decision Tree and Random Forest achieved malignant recall of 0.9286 (39 of 42 malignant samples detected). For a health-related classification task, malignant recall is an important metric to report alongside overall accuracy.

8.5 PCA comparison

Model	Original accuracy	PCA accuracy	Difference (percentage poin
Logistic Regression	98.25%	97.37%	-0.88
SVM	98.25%	97.37%	-0.88
Random Forest	95.61%	92.11%	-3.50
Decision Tree	91.23%	91.23%	+0.00

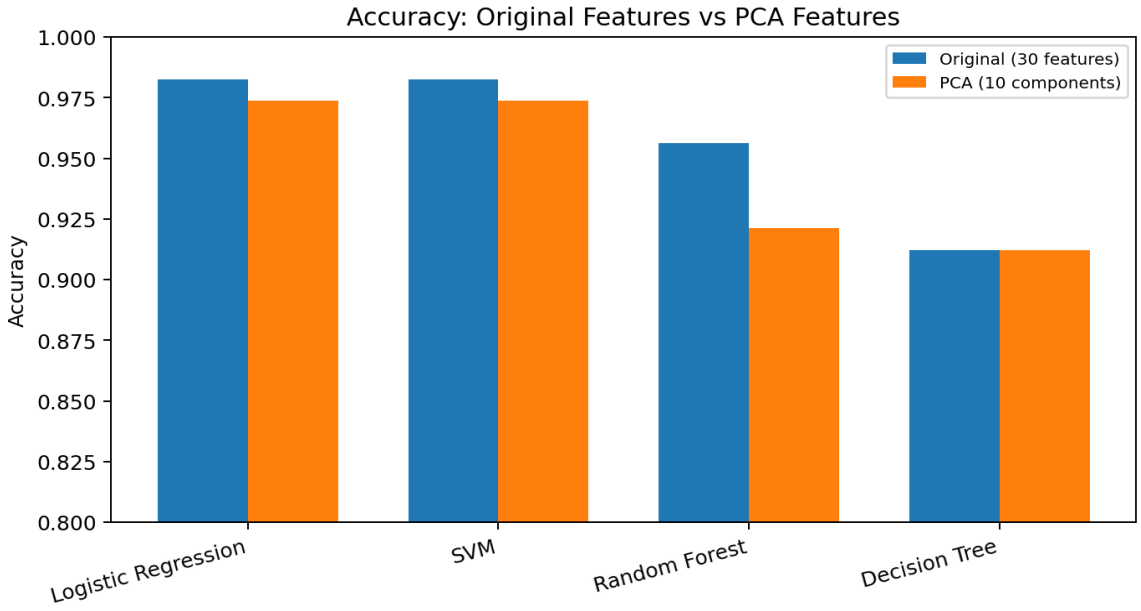


Figure 7. Accuracy comparison between the original feature set and the PCA representation.

PCA reduced 30 features to 10. Accuracy decreased from 98.25% to 97.37% for Logistic Regression and SVM, and from 95.61% to 92.11% for Random Forest. Decision Tree accuracy remained 91.23%. In this experiment, the original features gave better or equal accuracy, so they are retained as the main reported approach; PCA is presented as a dimensionality-reduction comparison.

8.6 Random Forest feature importance

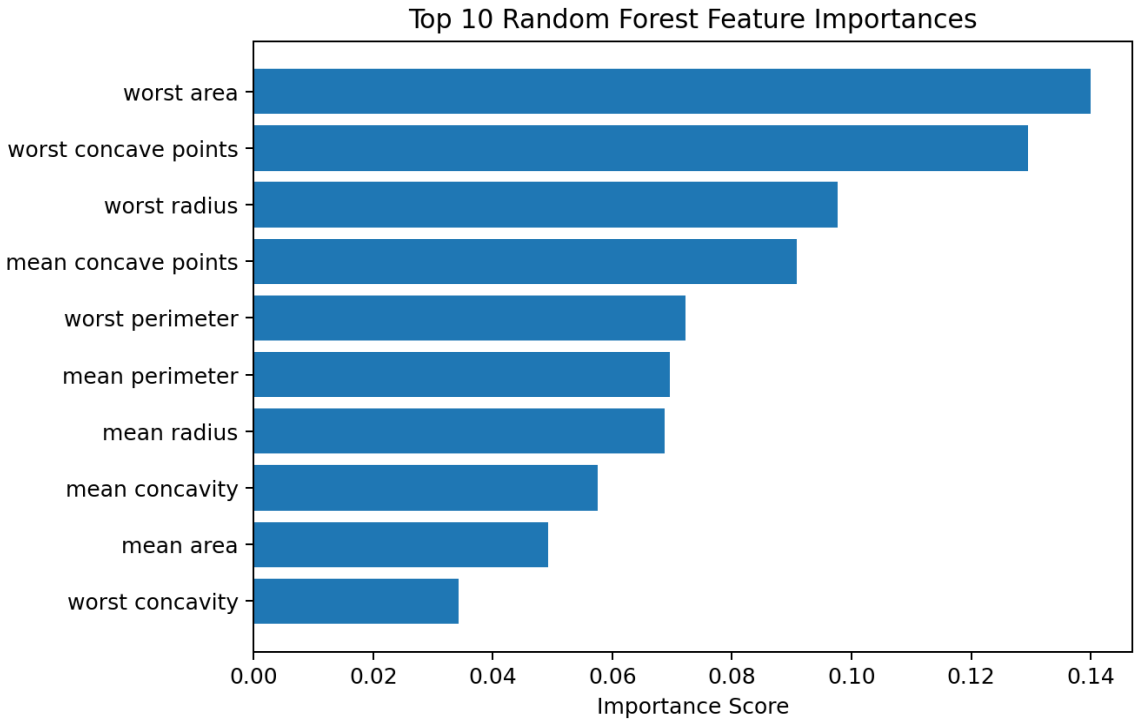


Figure 8. Top 10 features ranked by Random Forest feature importance.

Rank	Feature	Importance
1	worst area	0.1400
2	worst concave points	0.1295
3	worst radius	0.0977
4	mean concave points	0.0909
5	worst perimeter	0.0722
6	mean perimeter	0.0696
7	mean radius	0.0687
8	mean concavity	0.0576
9	mean area	0.0492
10	worst concavity	0.0343

Worst area had the highest importance score (0.1400), followed by worst concave points (0.1295) and worst radius (0.0977). Several size- and shape-related measurements appear in the top ten. Random Forest importance indicates how the trained model used these features; it does not establish causation.

9. Limitations and Future Scope

Limitations

• The evaluation uses a single stratified 80:20 split (114 test samples), so reported metrics may vary with a different split or dataset.

• The models were evaluated with fixed settings; systematic hyperparameter tuning and cross-validation were not reported in this notebook.

• No independent external dataset or prospective clinical validation was used.

• Feature importance from Random Forest is model-specific and should not be interpreted causally.
• The dataset and task are educational; the outputs must not be used for medical diagnosis or treatment decisions.

Future scope

• Use stratified cross-validation and report mean performance with variability.
• Tune model hyperparameters using cross-validation while keeping the final test set untouched.
• Evaluate calibration, threshold selection and malignant false-negative trade-offs.
• Validate on an independent dataset and assess generalization before considering any real-world application.
• Compare permutation importance or SHAP explanations with the current Random Forest importance ranking.

10. Conclusion

This case study compared Logistic Regression, SVM, Decision Tree and Random Forest on the Breast Cancer Wisconsin (Diagnostic) dataset. Using the original 30 standardized features, Logistic Regression and SVM achieved the highest test accuracy of 98.25%. Logistic Regression had the highest reported ROC-AUC at 0.9954. The malignant-specific evaluation showed that Logistic Regression and SVM each detected 41 of 42 malignant test samples. PCA reduced the feature space to 10 components while retaining 95.27% of variance, but did not improve test accuracy in this experiment. Overall, Logistic Regression using the original features is the best preliminary model among the evaluated settings on this split. Additional cross-validation and independent validation would be required before drawing broader conclusions.

References

1. Scikit-learn, “Breast cancer Wisconsin (diagnostic) dataset,” documentation for load_breast_cancer: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html
2. Google Colab notebook supplied by the project author, “Breast_Cancer_ML_Case_Study - Colab,” exported 10 October 2026.
3. Scikit-learn documentation for StandardScaler, PCA, LogisticRegression, SVC, DecisionTreeClassifier, RandomForestClassifier and classification metrics: https://scikit-learn.org/stable/